Far from the frontier: the China shock and firm innovation in Latin America

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Setting the scene

- China's weight as an exporter nearly doubled, 2002–2008: **4.1%** $\rightarrow$ **7.6%** of world exports
- This inflow of goods reached firms in **two distinct ways**:
  - as sellers, facing competition in their own market
  - as buyers, having more access to foreign inputs
- Theory says these two may pull in **opposite** directions, particularly for firms far from the frontier
- Question: for Latin American manufacturers, which effect was felt the strongest? Through which mechanisms?

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- Input access → innovation: usually **positive** (Ecuador: Bas and Paunov, 2021 ; India: (Bas and Berthou, 2016)), albeit sometimes weaker France: (Aghion et al., 2024)
- A single import-penetration measure **conflates both** — the estimated sign depends on which channel happens to dominate
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What this paper does

1. Decomposes trade data through input-output tables to create a **4-digit** industry exporting and importing country-industry dataset into competition and input-access exposure,
2. Builds import penetration and input access exposure measures for Argentina, Brazil and Mexico — **85%** of Latin American patenting
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Data

- **Patents** (PATSTAT): Panel of 2,576 innovating firms, 1996–2008
- **Trade:** multi-step estimation of imports of intermediates

- Industry i as **seller** $\rightarrow$ competition
  as **buyer** $\rightarrow$ input access

▸ Construction details

Data

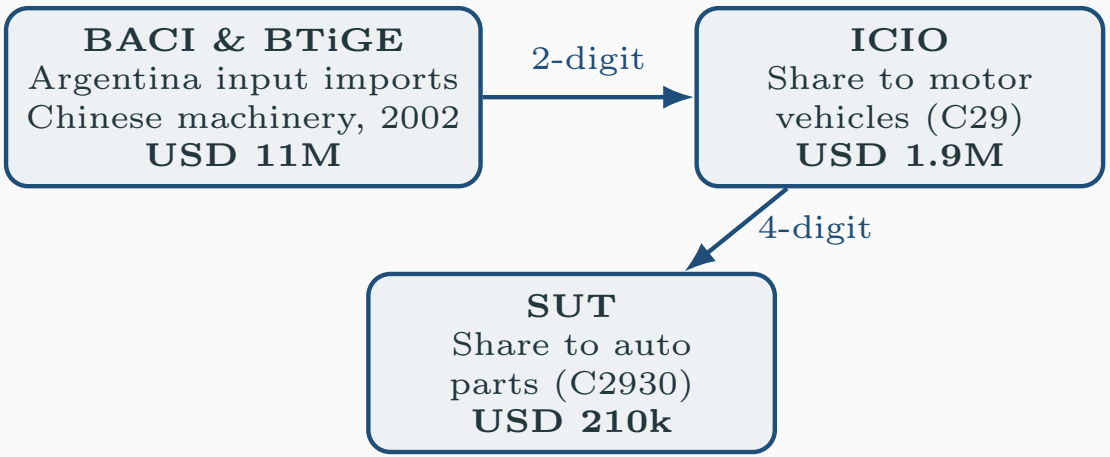
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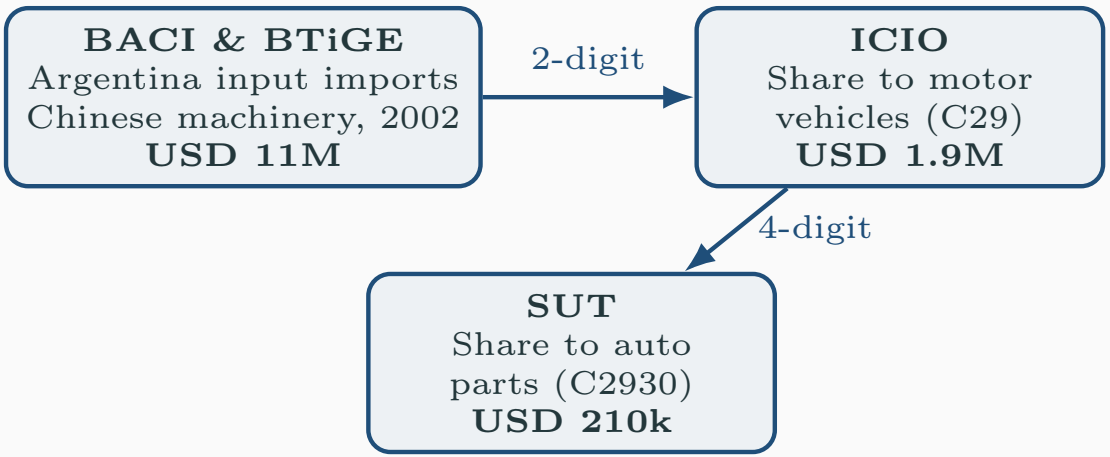


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Descriptives

- Patenting in Latin America is **highly concentrated:** Top 6% of innovators → 50% of granted patent families
- The vast majority (88%) of firms only innovated once between 1996–2008
- Most firms are from Brazil (77%) and are concentrated in four out of 23 2-digit sectors: machinery, chemicals, electrical equipment and other manufacturing (63%)

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Empirical strategy

$$\Delta Y_{f,c,i} = \beta_0 + \beta_1 \Delta IMP_{c,i}^{CHN} + \beta_2 \Delta IMPI_{c,i}^{CHN} + \delta_{c,g(i)} + \varepsilon_{f,c,i}$$

- Firm-level first difference: 1996–2001 vs. 2002–2008
- Outcome: DHS growth rate of patenting — defined even for firms with zero patents in a period
- Exposure: the rise of China as a trade partner (% of imports)
- Both measures are endogenous to domestic demand → instrumented with **China's rise in similar economies**
- Clustering at 4-digit industry level

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Results: input access dominates

	OLS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)
ΔIMP	-0.015^{**}		-0.016^{**}	-0.014^{*}		-0.020^{***}
	(0.007)		(0.007)	(0.008)		(0.007)
$\Delta IMPI$		0.016	0.025		0.117^{**}	0.100^{***}
		(0.025)	(0.025)		(0.050)	(0.028)
Num. obs.	2,576	2,576	2,576	2,576	2,576	2,576
Mean outcome	0.335	0.335	0.335	0.335	0.335	0.335
Country-sector FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.042	0.038	0.043	0.042	0.018	0.033
KP F				39.3	16.6	13.3

- +1pp Chinese competition share → −2% annual patenting
- +1pp Chinese input share → +10% annual patenting
- Joint estimation matters: the competition coefficient **grows** once input access is held constant

The effect concentrates in high-technology firms

	OLS		2SLS	
	Low tech	High tech	Low tech	High tech
ΔIMP	−0.014	−0.016	−0.012	−0.024**
	(0.009)	(0.011)	(0.010)	(0.011)
$\Delta IMPI$	−0.029	0.059**	0.013	0.135***
	(0.025)	(0.029)	(0.012)	(0.032)
Num. obs.	779	1,797	779	1,797
Mean outcome	0.228	0.382	0.228	0.382
Country-sector FE	Yes	Yes	Yes	Yes
R^2	0.073	0.038	0.069	0.030
KP F			36.9	11.9

- Both channels strongest among firms in **high-technology** sectors
- Difference between low- and high-tech firms is itself significant (interaction specification)

And runs through high-technology inputs

	OLS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta IMPI^{HT}$	0.038*	0.038**	0.047***	0.097***	0.061***	0.069***
	(0.020)	(0.017)	(0.015)	(0.034)	(0.021)	(0.020)
$\Delta IMPI^{LT}$		−0.091***	−0.077***		−0.144**	−0.090
		(0.032)	(0.026)		(0.069)	(0.059)
ΔIMP			−0.015***			−0.015***
			(0.004)			(0.005)
Num. obs.	2,576	2,576	2,576	2,576	2,576	2,576
Mean outcome	0.335	0.335	0.335	0.335	0.335	0.335
Country-sector FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.040	0.047	0.051	0.033	0.043	0.050
KP F				28.0	7.4	5.4

- High-tech coefficient positive and significant **throughout**
- Consistent with **learning**: firms gain from technology they don't already command, not simply cheaper inputs of any kind

Robustness

- **Alternative outcomes:** raw patent-count differences, patent-stock weighting, always-patenting subsample (intensive margin only) — input access remains positive & significant throughout ▸ Table A.9
- **Lawyer-effect:** Citations per patent rules out defensive patenting ▸ Table A.8
- **Alternative IO source:** World Input-Output Database in place of ICIO — coefficient weaker but same sign, on the Brazil/Mexico subsample ▸ Table A.11
- **Alternative donor pool:** 6 non-LatAm emerging economies — consistent sign/magnitude, weaker first stage ▸ Table A.12
- **Reduced form:** instrument enters positively/significantly for input access; competition instrument weaker (10% level) ▸ Table A.10

Main takeaways

- Two channels, opposite signs: competition discourages innovation, input access enables it by a much larger magnitude
- Net effect of the China shock on Latin American patenting: **positive**, driven by the input access side
- Reconciles the literature: a bundled import-penetration measure nets out whichever channel dominates a country's trade structure
- For nonfrontier countries, the influx of Chinese goods, even in the early 2000s, already provided a learning experience for firms to innovate

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Thank you

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Data — construction detail

Patents: Granted applications (INPADOC families, which avoid double counting). Information on location, firm name, filing date, number of citations, technology classification.

Name-matching: 3,631 raw strings → 2,923 firms (regex + Jaro–Winkler) → 2,576 with assignable sector

Industry assignment: $\theta_{f,i} = n_{f,i} / \sum_k n_{f,k}$, assigned to firm's max-weight industry (1980–2007 window)

Exposure measures:

$$IMP^{CHN}_{c,i,t} = EXP^{FD}_{x,c,i,t} + EXP^{INP}_{x,c,i,t} \qquad IMPI^{CHN}_{c,i,t} = \sum_j EXP_{x,c,j,i,t}$$

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Data — Sample distribution

Variable	Mean	SD	1st	10th	25th	50th	75th	90th	99th	99.9th
Pat	3.3	14.4	1.0	1.0	1.0	1.0	2.0	5.0	34.5	144.0
ΔPat	0.3	1.9	−2.0	−2.0	−2.0	2.0	2.0	2.0	2.0	2.0
Cit	25.0	176.7	0.0	0.0	0.0	0.0	10.0	38.0	433.8	2620.8
ΔCit	0.2	1.2	−2.0	−2.0	0.0	0.0	1.3	2.0	2.0	2.0
ΔIMP	12.3	10.6	−0.3	2.3	3.9	8.9	16.3	29.6	40.3	43.7
$\Delta IMPI$	7.6	3.6	−4.3	3.5	5.5	7.8	9.2	10.8	15.9	30.4

Deltas for Pat and Cit follow the DHS growth-rate specification (bounded $[-2, 2]$); ΔIMP and $\Delta IMPI$ are percentage-point changes in China's partner share, 2002–2008.

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Empirical strategy annex

The DHS growth rate:

$$\Delta Y_{f,c,i} = 2 \cdot \frac{\bar{p}_{f,c,i,t} - \bar{p}_{f,c,i,t-1}}{\bar{p}_{f,c,i,t} + \bar{p}_{f,c,i,t-1}}$$

Instrument:

$$\Delta flow^{CHN}_{O(c),i} = \frac{\sum_{d \in O(c)} flow^{CHN}_{d,i,2008}}{\sum_{d \in O(c)} flow^{World}_{d,i,2008}} - \frac{\sum_{d \in O(c)} flow^{CHN}_{d,i,2002}}{\sum_{d \in O(c)} flow^{World}_{d,i,2002}}$$

- Competition instrument: leave-one-out pool of 8 Latin American economies (AR, BR, CL, CO, MX, PA, PE, UY)
- Input-access instrument: rest-of-world aggregate excluding China and the treated country — no IO structure available for smaller LatAm donors

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Alternative outcome definitions

	Patent counts			Patent-weighted			Always-patenting		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
ΔIMP	−0.002		−0.004*	−0.014*		−0.022***	−0.011*		−0.014**
	(0.002)		(0.002)	(0.008)		(0.007)	(0.006)		(0.006)
$\Delta IMPI$		0.036***	0.033***		0.138**	0.115***		0.066**	0.043*
		(0.010)	(0.007)		(0.054)	(0.025)		(0.030)	(0.024)
Num. obs.	2,576	2,576	2,576	2,576	2,576	2,576	314	314	314
Mean outcome	0.058	0.058	0.058	0.347	0.347	0.347	−0.019	−0.019	−0.019
Country-sector FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.009	0.012	0.012	0.044	0.015	0.035	0.108	0.109	0.114
KP F	39.3	16.6	13.3	57.4	13.3	18.5	29.5	11.5	16.6

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Patents vs. citations per patent (lawyer effect)

	Patents			Citations per patent		
	(1)	(2)	(3)	(4)	(5)	(6)
ΔIMP	-0.014^{*}		-0.023^{***}	-0.000		-0.004
	(0.008)		(0.008)	(0.004)		(0.003)
$\Delta IMPI$		0.165^{*}	0.131^{***}		0.054^{**}	0.048^{**}
		(0.083)	(0.032)		(0.026)	(0.020)
Num. obs.	1,842	1,842	1,842	1,842	1,842	1,842
Mean outcome	0.260	0.260	0.260	0.230	0.230	0.230
Country-sector FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.038	0.047	0.049	0.009	0.013	0.013
KP F	34.9	10.9	12.9	34.9	10.9	12.9

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Alternative standard-error clustering

	ISIC 2-digit			Country-ISIC 4-digit		
	(1)	(2)	(3)	(4)	(5)	(6)
ΔIMP	-0.014^{*}		-0.020^{***}	-0.014^{*}		-0.020^{***}
	(0.007)		(0.006)	(0.007)		(0.007)
$\Delta IMPI$		0.117^{**}	0.100^{***}		0.117^{**}	0.100^{***}
		(0.049)	(0.025)		(0.047)	(0.029)
Num. obs.	2,576	2,576	2,576	2,576	2,576	2,576
Mean outcome	0.335	0.335	0.335	0.335	0.335	0.335
Country-sector FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.040	0.047	0.048	0.040	0.047	0.048
KP F	38.9	16.4	12.9	38.9	16.4	12.9

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Reduced-form estimates

	2SLS			Reduced-form OLS		
	(1)	(2)	(3)	(4)	(5)	(6)
ΔIMP	−0.007			−0.020**		
	(0.012)			(0.008)		
$\Delta IMPI$		0.117**			0.100***	
		(0.050)			(0.030)	
Z^{IMP}			−0.018			−0.013*
			(0.011)			(0.007)
Z^{IMPI}			0.158***			0.147***
			(0.042)			(0.036)
Num. obs.	2,576	2,576	2,576	2,576	2,576	2,576
Mean outcome	0.335	0.335	0.335	0.335	0.335	0.335
Country-sector FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.038	0.047	0.049	0.040	0.047	0.048
KP F	38.9	16.4	12.9			

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Alternative input–output source: WIOD vs. ICIO

	ICIO		WIOD	
	(1)	(2)	(3)	(4)
$\Delta IMPI$	0.101**	0.084***	0.069*	0.076**
	(0.050)	(0.029)	(0.036)	(0.035)
ΔIMP		−0.018**		−0.017**
		(0.007)		(0.007)
Num. obs.	2,233	2,233	2,233	2,233
Mean outcome	0.428	0.428	0.428	0.428
Country-sector FE	Yes	Yes	Yes	Yes
R^2	0.003	0.017	0.013	0.019
KP F	14.6	17.6	54.0	24.9

Sample restricted to Brazil & Mexico (WIOD lacks Argentina coverage).

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Alternative donor pool: non-Latin American economies

	Baseline basket			Alternative basket		
	(1)	(2)	(3)	(4)	(5)	(6)
ΔIMP	-0.014^{*}		-0.020^{***}	-0.017^{*}		-0.020^{***}
	(0.008)		(0.007)	(0.010)		(0.007)
$\Delta IMPI$		0.117^{**}	0.100^{***}		0.080^{*}	0.125^{***}
		(0.050)	(0.028)		(0.042)	(0.042)
Num. obs.	2,576	2,576	2,576	2,576	2,576	2,576
Mean outcome	0.335	0.335	0.335	0.335	0.335	0.335
Country-sector FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.042	0.018	0.033	0.035	0.030	0.034
KP F	39.3	16.6	13.3	33.9	11.2	5.8

Alternative basket: India, Indonesia, South Africa, Thailand, Turkey, Vietnam.